

LAB # 3**DECLARING VARIABLES AND TO TAKE INPUT FROM USER****Lab Tasks**

- 1- Write a program that take four floating numbers from keyboard and prints their sum, product and average.
- 2- Write a program to calculate the area of the circle, taking the value of the radius from the user.
- 3- Write a program that prints to calculate the age in days by using the formula:
days = years * 365.
- 4- Write a C++ program to convert temperature from degree Celsius to Fahrenheit.

/* celsius to fahrenheit conversion formula */

fahrenheit = (celsius * 9 / 5) + 32;

- 5- Mention output for following code fragment.

```
cout << "Size of char : " << sizeof(char) << endl;
cout << "Size of int : " << sizeof(int) << endl;
cout << "Size of short int : " << sizeof(short int) << endl;
cout << "Size of long int : " << sizeof(long int) << endl;
cout << "Size of float : " << sizeof(float) << endl;
cout << "Size of double : " << sizeof(double) << endl;
cout << "Size of wchar_t : " << sizeof(wchar_t) << endl;
```

- 6- State the order of evaluation the operation in each of the following C++ statements, and show the value of x after each statement is performed.

a) $x = 7 + 3 * 6 / 2 - 1;$

b) $x = 2 \% 2 + 2 * 2 - 2 / 2;$

- 7- Write equivalent C++ statements for given mathematical expressions

a)
$$\text{root} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

b)
$$z = \frac{x+y}{x-y}$$

- 1- Write a program that take four floating numbers from keyboard and prints their sum, product and average.

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n";
6      float num1,num2,num3,num4;
7      cout << "Enter four number: \n";
8
9      cin >> num1>>num2>>num3>>num4;
10
11     float sum = num1+num2+num3+num4;
12     float product = num1 * num2 * num3 * num4;
13     float average = (num1 + num2 + num3 + num4)/4;
14
15     cout <<"Sum: " << sum << endl;
16     cout << "Product: " << product << endl;
17     cout <<"Average: " << average << endl;
18
19     return 0;
20 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
Enter four number:
2
4
3
5
Sum: 14
Product: 120
Average: 3.5
```

- 2- Write a program to calculate the area of the circle, taking the value of the radius from the user

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6      float radius, pi;
7      pi = 3.142;
8
9      cout << "Enter radius: ";
10     cin >> radius;
11     cout << "Area of Circle: "<< pi*(radius*radius);
12
13     return 0;
14 }
15
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

Enter radius: 8
Area of Circle: 201.088

Output
C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 3492) exited with code 0 (0x0).
Show output from To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automaticall

- 3- Write a program that prints to calculate the age in days by using the formula:
 $\text{days} = \text{years} * 365$.

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6      float age;
7      cout << "Enter age : ";
8      cin >> age;
9      cout << "Age in days : "<< age * 365;
10     return 0;
11 }
12
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

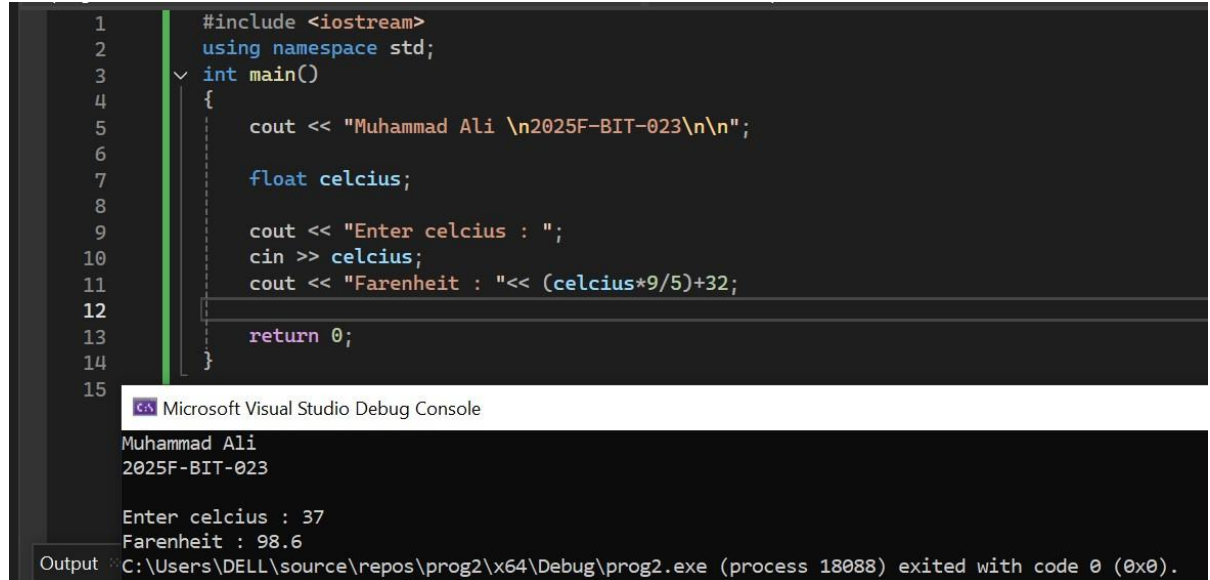
Enter age : 18
Age in days : 6570

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 3104) exit
Press any key to close this window . . .

- 4- Write a C++ program to convert temperature from degree Celsius to Fahrenheit.

/* celsius to fahrenheit conversion formula */

fahrenheit = (celsius * 9 / 5) + 32;



```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      float celcius;
8
9      cout << "Enter celcius : ";
10     cin >> celcius;
11     cout << "Farenheit : "<< (celcius*9/5)+32;
12
13     return 0;
14 }
15
```

Microsoft Visual Studio Debug Console

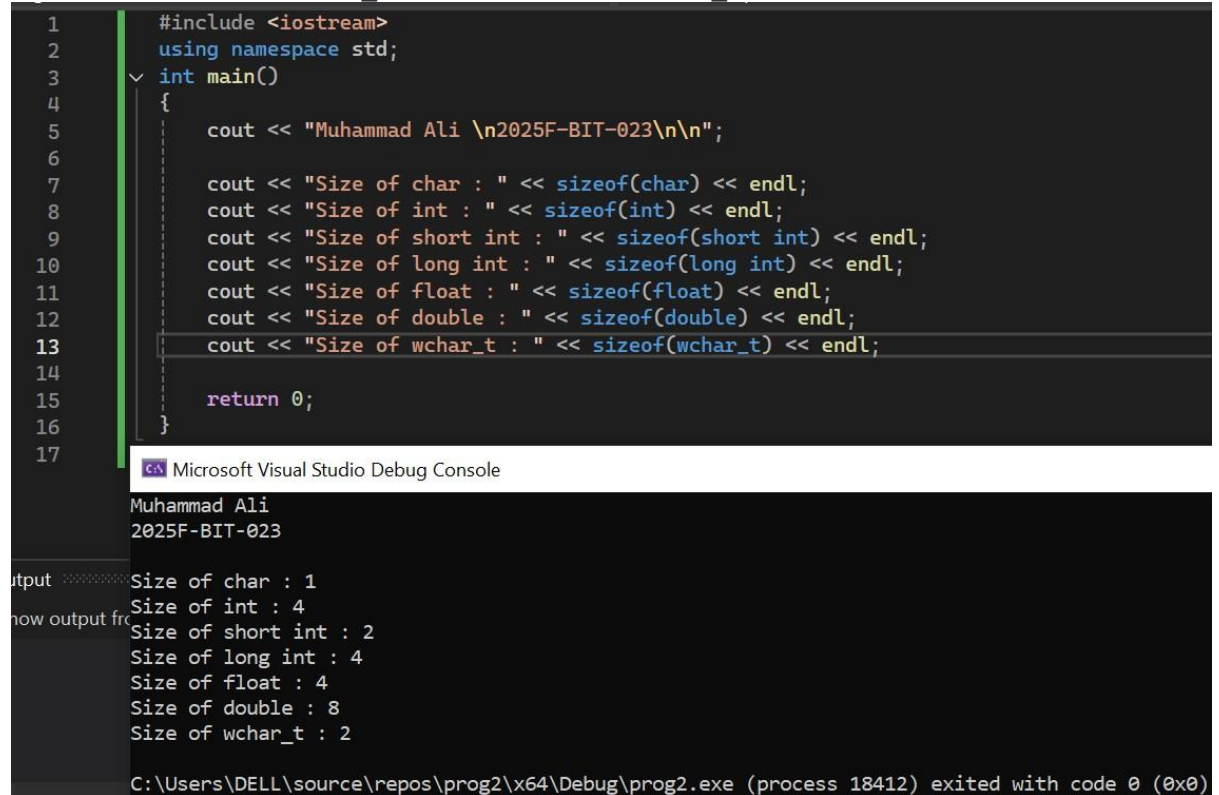
Muhammad Ali
2025F-BIT-023

Enter celcius : 37
Farenheit : 98.6

Output C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 18088) exited with code 0 (0x0).

5- Mention output for following code fragment.

```
cout << "Size of char : " << sizeof(char) << endl;
cout << "Size of int : " << sizeof(int) << endl;
cout << "Size of short int : " << sizeof(short int) << endl;
cout << "Size of long int : " << sizeof(long int) << endl;
cout << "Size of float : " << sizeof(float) << endl;
cout << "Size of double : " << sizeof(double) << endl;
cout << "Size of wchar_t : " << sizeof(wchar_t) << endl;
```



The screenshot displays a C++ program in Visual Studio Code. The code defines a `main` function that prints a name and ID, followed by the sizes of various data types using `sizeof`. The output window shows the execution results, including the printed name, ID, and the sizes of `char`, `int`, `short int`, `long int`, `float`, `double`, and `wchar_t`.

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      cout << "Size of char : " << sizeof(char) << endl;
8      cout << "Size of int : " << sizeof(int) << endl;
9      cout << "Size of short int : " << sizeof(short int) << endl;
10     cout << "Size of long int : " << sizeof(long int) << endl;
11     cout << "Size of float : " << sizeof(float) << endl;
12     cout << "Size of double : " << sizeof(double) << endl;
13     cout << "Size of wchar_t : " << sizeof(wchar_t) << endl;
14
15     return 0;
16 }
17
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

Size of char : 1
Size of int : 4
Size of short int : 2
Size of long int : 4
Size of float : 4
Size of double : 8
Size of wchar_t : 2

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 18412) exited with code 0 (0x0)

6- State the order of evaluation the operation in each of the following C++ statements, and show the value of x after each statement is performed.

a) $x = 7 + 3 * 6 / 2 - 1;$

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      cout << "x = 7 + 3 * 6 / 2 - 1\n";
8      cout << "x = 7 + 3 * 3 - 1\n";
9      cout << "x = 7 + 9 - 1\n";
10     cout << "x = 16 - 1\n";
11     cout << "x = 15\n";
12
13     return 0;
14 }
15
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023

x = 7 + 3 * 6 / 2 - 1
x = 7 + 3 * 3 - 1
x = 7 + 9 - 1
x = 16 - 1
x = 15

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 7572) exited v
```

b) $x = 2 \% 2 + 2 * 2 - 2 / 2;$

```

1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      cout << "x = 2 % 2 + 2 * 2 - 2 / 2 \n";
8      cout << "x = 0 + 2 * 2 - 2/2 \n";
9      cout << "x = 0 + 2 * 2 - 1 \n";
10     cout << "x = 0 + 4 - 1 \n";
11     cout << "x = 4 - 1 \n";
12     cout << "x = 3 \n";
13
14     return 0;
15 }
16
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023
x = 2 % 2 + 2 * 2 - 2 / 2
x = 0 + 2 * 2 - 2/2
x = 0 + 2 * 2 - 1
x = 0 + 4 - 1
x = 4 - 1
x = 3

```

7-Write equivalent C++ statements for given mathematical expressions

a) root = $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

```

1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      cout << "[-b + sqrt(b*b - 4*a*c) / 2*a] \n";
8
9
10     return 0;
11 }
12
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023
[-b + sqrt(b*b - 4*a*c) / 2*a]
C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 4632) exited with co
Press any key to close this window . . .

```

b) $z = \frac{x+y}{x-y}$

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali \n2025F-BIT-023\n\n";
6
7      cout << "z = x + y / x - y \n";
8
9
10     return 0;
11 }
12
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

z = x + y / x - y

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 10736) exited wi